

Neuro-CIM: ADC-Less Neuromorphic Computing-in-Memory Processor With Operation Gating/Stopping and Digital–Analog Networks

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Abstract—A highly energy-efficient neuromorphic computing-in-memory (Neuro-CIM) processor is proposed for ultralow-power deep learning applications. Neuro-CIM can support spiking neural network (SNN) to eliminate the power and area overhead of previous CIM processor. The sign extended bits gating reduces the bitline (BL) voltage switching rate due to negative small-magnitude weights allowing 38% power reduction at 8-b weight condition and 25% at 4-b weight condition. In addition, Neuro-CIM replaces high-precision analog-to-digital converter (ADC) with 1-b comparator by exploiting the characteristic of the SNN, and thus, power and area efficiencies are significantly enhanced. Furthermore, the early stopping scheme terminates unnecessary neuronal operations, reducing power consumption by 31%. In addition, the analog and digital networks are integrated for high reconfigurability and energy efficiency. The analog network with voltage folding circuit enables accurate analog-domain aggregation by increasing the dynamic range without compromising the voltage resolution. The digital network-in-memory supports input–output channel extension for high reconfigurability and input data reuse scheme for reducing input memory (IMEM) access. Neuro-CIM is fabricated in 28-nm CMOS technology and occupies the 2.9-mm² die area. It achieves the state-of-the-art energy consumption per classification of 0.72 μ J and 92.1% accuracy for CIFAR-10 with 4-b input and 4-b weight and 372.2 μ J and 65.8% accuracy for ImageNet with 6-b input and 8-b weight at 200 MHz, and 1.1-V conditions. Moreover, the proposed CIM processor achieves 310.37 tera operations per second/watt (TOPS/W) and 90.7% accuracy with 4-b input and 1-b weight for Canadian Institute for Advanced Research, 10 classes (CIFAR-10) classification.

Index Terms—Computing-in-memory (CIM), deep learning ASIC, neuromorphic computing, spiking neural network (SNN).

I. INTRODUCTION

RECENTLY, the computing-in-memory (CIM) processors [1], [2], [3], [4], [5], [6] are proposed to remove

Manuscript received 28 December 2022; revised 15 March 2023; accepted 27 April 2023. Date of publication 19 May 2023; date of current version 26 September 2023. This article was approved by Associate Editor Sanu K. Mathew. This work was supported by Samsung Electronics Company Ltd., Seoul, South Korea, under Grant IO201207-07799-01. (Corresponding author: Hoi-Jun Yoo.)

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Color versions of one or more figures in this article are available at <https://doi.org/10.1109/JSSC.2023.3273238>.

Digital Object Identifier 10.1109/JSSC.2023.3273238

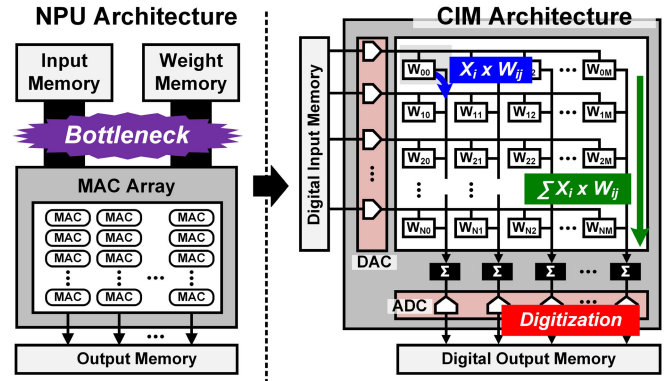


Fig. 1. Motivation of CIM architecture.

the overhead of previous neural processing unit (NPU) architecture based on von Neumann architecture. The neural network processing requires large memory access for input and weight, and it causes memory bottleneck, which reduces the performance of NPU, as shown in Fig. 1. The CIM performs multiple multiplications between weight and activation simultaneously and accumulates them in the current domain. The bitline (BL) voltage represents the multiply-and-accumulation (MAC) results, and the result is transferred to the digital processing unit or memory by the analog-to-digital converter (ADC). By using the CIM architecture, the number of memory accesses and power consumption can be reduced, thus solving the memory bottleneck. However, the CIM still has a major challenge, as the ADC causes large power and area overhead. In the CIM operation, multiple word lines (WLs) are activated simultaneously, and the BL voltage (V_{BL}) swing is limited by the power supply voltage, signal margin, and linearity. Therefore, it affects ADC resolution, and high-precision ADC is required for accurate computing.

The previous CIM processor [1] increased the number of WLs for a higher parallelism. It enabled the 1152 WLs simultaneously and performed MACs in BL. Then, the ADC generated the digital result from the V_{BL} . To achieve high accuracy for CIFAR-10 classification, high-precision ADC (8 b) was used, and it consumed 59% of total power, as shown in Fig. 2. Moreover, higher precision ADC should be required for complex task, such as ImageNet classification,

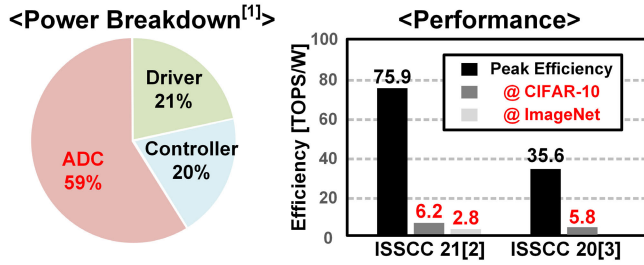


Fig. 2. Overhead of previous CIM processors.

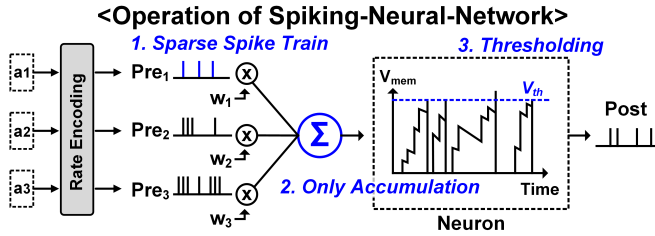


Fig. 3. Operation of SNN.

and it will cause larger power overhead. In order to reduce this overhead and increase performance, sparsity handling CIM processors [2], [3], [4] have been proposed. Previous processor [2], [3] skipped the unnecessary operations by blockwise sparsity-optimized CIM macro to achieve high energy efficiency. Also, the other processor [4] skipped the ADC operation for negative outputs by using rectified linear unit (ReLU)-based skipping. These methods could achieve high maximum (MAX) energy efficiency at ideal condition (90% sparsity), but large performance degradation occurs in real world, as shown in Fig. 2. The sparsity of the networks for real-world tasks is not very high ($\sim 50\%$) [7], and the efficiency decreases by $\sim 90\%$ compared with their ideal computational scenario.

In this article, we propose neuromorphic CIM (Neuro-CIM) architecture that achieves high energy efficiency regardless of the sparsity conditions. Neuromorphic computing can achieve low-power consumption by mimicking the brain, as it exploits an event-driven characteristic. A spiking neural network (SNN) is a type of neuromorphic network that models the behavior of biological neurons more closely than other types. In an SNN, neurons communicate with each other by sending discrete, binary “spike” signals rather than continuous analog signals. During inference in an SNN, the network receives an input signal, which is typically encoded as a sequence of spikes. This input signal is propagated through the network, with each neuron receiving input from other neurons and potentially generating spikes of its own. The propagation of spikes through the network can be modeled using a variety of different algorithms, but one common approach is to use a time-based encoding scheme in which spikes are assigned a specific time value. In this scheme, the input signal is converted into a series of spikes, with each spike representing a particular feature of the input. As the input spikes propagate through the network, they trigger the firing of other neurons, which, in turn, generate spikes of their own. This process continues until the output of the network is computed, typically

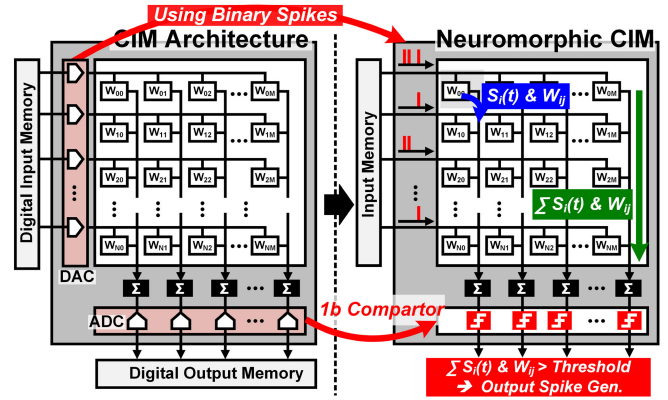


Fig. 4. Motivation of Neuro-CIM architecture.

by reading out the spikes generated by a particular output neuron or group of neurons. In summary, during inference in an SNN, an input signal is encoded as a sequence of spikes, which are then propagated through the network using a time-based encoding scheme. The network responds to the input by generating spikes of its own, which ultimately lead to the computation of the network’s output. Fig. 3 shows the operation of each layer in SNN that performs time-serial accumulations instead of MACs in convolutional neural network (CNN). In addition, rate encoding is used to generate spike trains, and thus, it fires more spikes at larger input feature. When a spike is delivered to the neuron, the weight is added to the membrane potential. The membrane potential and the threshold are compared at every time step. Only if the potential is larger than the threshold, an output spike is generated, and the neuron’s potential is reset. Therefore, the spike trains generated by encoding or neuronal operations are very sparse, and SNN shows a very small amount of computation due to sparse spike trains. In summary, the SNN is suitable for low power, because it has an event-driven characteristic and requires low-cost accumulation instead of high-cost MAC [8].

The Neuro-CIM processor (Fig. 4) has two advantages compared to the previous CIM. First, it can guarantee high input sparsity due to the event-driven operation of neuromorphic network. The input data of CNN are replaced with the sparse spike train by encoding, and the energy efficiency increases, because WL is enabled only when the spike occurs. Fig. 5 compares the number of operations in CNN and SNN, and the number of WL driving of SNN decreases by 74% compared with CNN. In addition, the high-precision ADC can be replaced with a 1-b comparator, because the output spike is generated by a simple comparison between the membrane potential and threshold. Therefore, it can reduce the power and area overhead.

However, some challenges need to be resolved to implement a low-power Neuro-CIM processor. Fig. 6 shows the voltage range limitation due to multiple WL driving. The WL is driven by an input spike at each cycle, and the current of compute BL and the voltage of the capacitor node increase according to the summation of logical multiplication results. The maximum voltage of capacitor node is limited by VBL and MAX. Also, it is not large enough due to factors, such as power supply voltage, signal margin, and linearity.

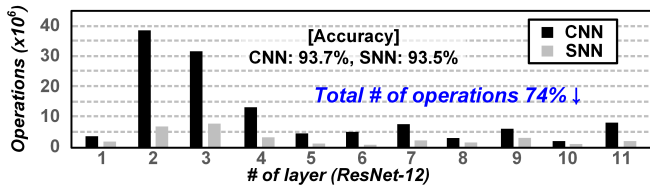


Fig. 5. Efficiency improvement of SNN (@ResNet-12, CIFAR-10).

For highly accurate CIM, high-precision ADC is required, and it requires large power and area. Also, if we increase the number of WLs for high parallelism, the resolution of ADC becomes too small to achieve high accuracy. Moreover, it still has insufficient sparsity in real conditions to reduce power overhead. Each cell stores the weight data, and many nonzero data cause the BL current rising at CIM operation. Therefore, large computation power is consumed due to dynamic BL voltage switching. In addition, many people may think the accuracy of the neuromorphic system is not good enough. However, CNN-to-SNN conversion [9], [10], [11], [12] was proposed to solve the accuracy limitation of spike time-dependent plasticity (STDP) algorithm [13], [14], [15], [16], which is the conventional training method for SNN. For network conversion, CNN training is performed first, and then, the trained parameters are shared with the SNN. In this process, additional retraining is not required, and only finding the threshold value of each neuron is required. Since threshold determination is performed only once in the beginning, there is no additional overhead at run time. Moreover, the converted SNN (CIFAR-10: 93.4%, Canadian Institute for Advanced Research, 100 classes (CIFAR-100): 69.7%, and ImageNet: 69.0%) shows almost the same accuracy as CNN [12] (CIFAR-10: 93.7%, CIFAR-100: 71.8%, and ImageNet: 70.0%). Therefore, it is possible to overcome the accuracy limitation and design a highly energy-efficient CIM processor.

To solve the challenges and achieve high energy efficiency, this article proposes Neuro-CIM with five key features: 1) sign extended bits gating reduces the V_{BL} switching for power reduction; 2) ADC-less neuron operation eliminates the area and power overhead of high-precision ADC; 3) early stopping reduces the V_{BL} switching by terminating unnecessary neuron operations; 4) analog network-in-memory with voltage folding extends the voltage range for high accuracy, and it can support low-power multimacro aggregation for high scalability; and 5) digital network-in-memory is integrated for high reconfigurability, and input reuse scheme can reduce the redundant memory access for low-power consumption.

II. OVERALL ARCHITECTURE

Fig. 7 shows the overall architecture of Neuro-CIM, which integrates 16-kB input memory (IMEM), 16-kB output memory (OMEM), 16 CIM macros, and analog/digital network-in-memory. Each macro integrates 16 banks and stores 1.25-kB weight. The digital network transmits the data of the IMEM to each bank, and the output generated by the bank is transmitted to the OMEM or other banks. Each bank integrates a 64×10 8T cell array, spike encoder, input buffer, capacitor-digital-to-analog converter (DAC) multibit adder,

voltage folding-and-count logic, firing logic, early stopping logic, and programmable gain amplifier (PGA). Data contained in the input buffer are transferred to the spike encoder, and the generated spikes are used to drive each WL. We use spike encoding, which was proposed in [17] to reduce the delay and energy overhead due to encoding. Also, the architecture of spike encoder was described in [18]. One sub-WL driver is connected to five cells, and early stopping is supported by disabling the sub-WL driver for power reduction. The enable signal for early stopping is generated by the early stopping logic by comparing the membrane voltage with a threshold for early stopping. Also, the firing logic generates output spikes by comparing the V_{BL} and threshold of the neuron, and the number of output spikes recorded by counters. Each bank has two “-1” flag BLs for sign extended bits gating, and power reduction is achieved by reducing the amount of charge accumulated in the BL capacitor by these new flag BLs.

In Neuro-CIM, we added two P-MOSFETs to the basic 6T cell. By using 8T SRAM, MB/ μm^2 is reduced by 47% compared with standard 6T SRAM. The BL is precharged to the GND, and whenever a read occurs, the BL capacitor is charged to compute the membrane potential. Until it reaches the threshold, the charge is accumulated in capacitor continuously. The analog network connects the BLs of banks in different macros to increase the macro size for high scalability. By connecting BLs, aggregation is completed by charge sharing between two different BL capacitors. At this time, PGA and voltage folding-and-count logic are used to maintain the resolution (V_{LSB}). PGA consists of a four amplifiers and eliminates the reduction of V_{LSB} caused by charge sharing. The input of PGA is the BL voltage of each bank, and the controller determines how much gain to use in the PGA. The amplified voltage is transmitted to other bank, and the average value is obtained through charge sharing for aggregation. When charge sharing is performed, a decrease in V_{LSB} occurs, but PGA prevents this by amplifying each BL voltage just before charge sharing.

III. PROPOSED NEURO-CIM PROCESSOR

A. Sign Extended Bits Gating

Fig. 8 shows the weight distribution of the neural network. ResNet-12 for CIFAR-10 requires 8-bit weights following a Gaussian distribution to perform inference without accuracy loss. At this time, most of the positive values are located near zero, and the upper 4 b has mostly “0000.” In addition, most of the negative values are also located near zero, and the upper 4 bit mostly has “1111.” The graph at the right of Fig. 8 shows the ratio of the upper 4 b (MSB) and the lower 4 b (LSB) having “0000” or “1111” among the total weights of each layer (@ResNet-12, CIFAR-10). In this case, more than 95% of the MSB is “0000” or “1111.” Due to these characteristics, the previous CIM architecture consumes redundant power. When the multiple WLs are enabled by inputs, multiple accumulations are performed in BL, and the V_{BL} must increase proportionally to the number of “1’s” in the cells. Therefore, the large increase in V_{BL} occurs due to many consecutive 1’s (“1111”) for upper BLs, and the CIM processor must consume large power. To solve this problem,

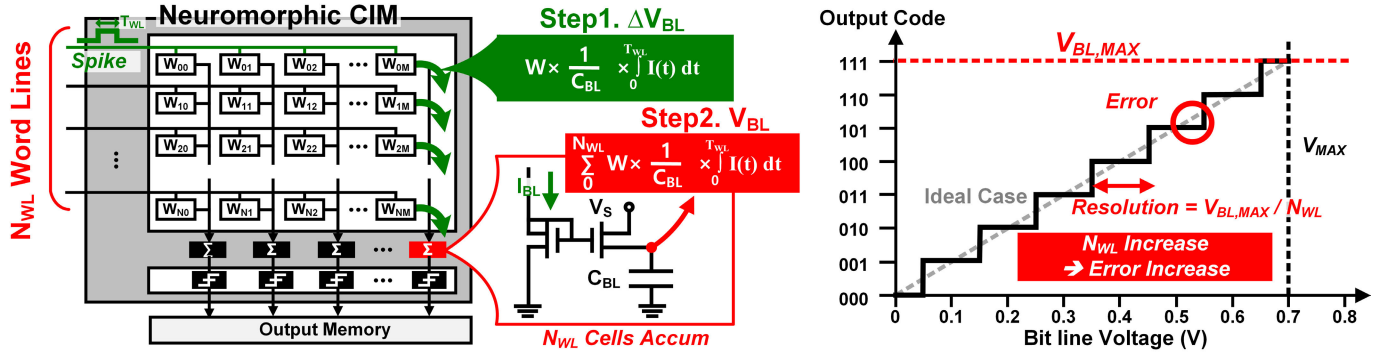


Fig. 6. Limitation of Neuro-CIM architecture.

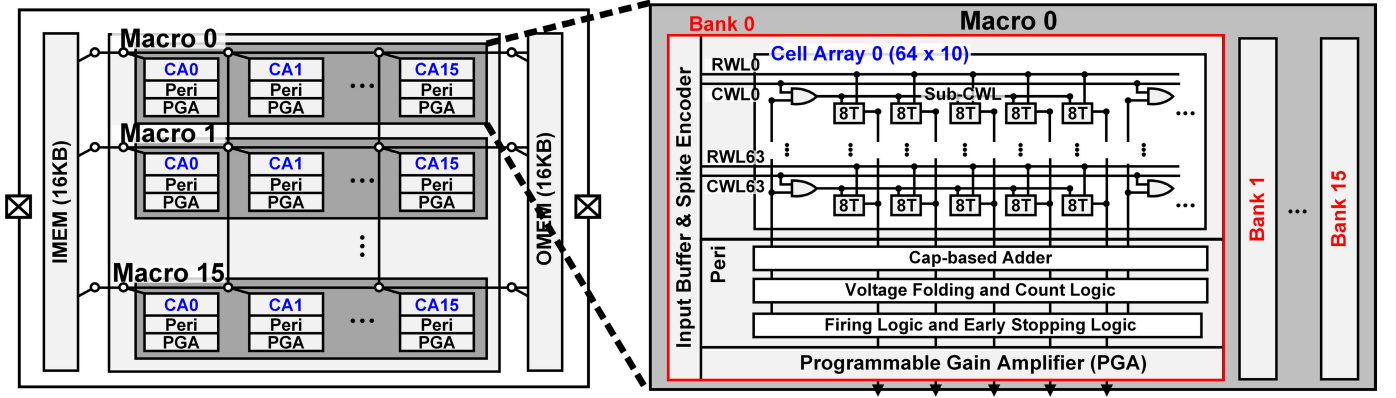


Fig. 7. Overall architecture of Neuro-CIM.

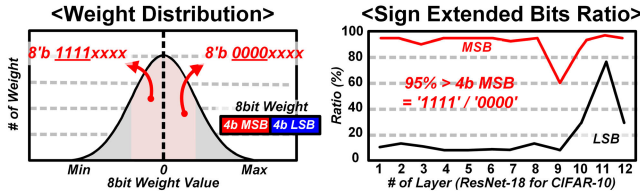


Fig. 8. Motivation of sign extended bits gating.

we propose flag BLs and sign extended bits gating to reduce the power consumption by consecutive 1's.

As described in Fig. 9, there are two “−1” flag BLs to indicate that the data stored in the next consecutive four BLs are all “1.” When the upper 4-b data are “1111,” “1” is stored in the first “−1” flag cell, and the upper 4-b data are replaced with “0000.” When the both upper 4 bit and the lower 4 b are “1111,” “1” is stored in the both two “−1” flag cells, and the weight data are replaced with “00000000” and stored. Afterward, when the CIM operation is performed, the voltage rise of the upper BLs is greatly reduced, because a lot of “1’s” existing in the upper four columns are replaced with “0’s.” Fig. 10 shows the operation with 4-b weights. In the case of 4-bit weight, the MSB corresponds to the upper 2 bits. When the upper 2 b is “11,” “1” is stored in the “−1” flag cell. Also, in the case of LSB, bits gating is not applied unlike the 8-b mode, because the probability that 2-b LSB is “11” is very low in the case of 4 b. As a result, power consumption of CIM macro is reduced by 38% for 8-b weight and by 25% for 4-b weight.

B. ADC-Less Neuron Operation

Fig. 11 shows the multibit operation with 8-b weight to generate complete membrane potential. In this case, the negative value of threshold ($-1 \times \text{threshold}$) and weights are stored together in cell array, and operation is performed. At the first time step, the WLs of the rows where the spike occurs are turned on. Also, the WL of the row, which stores $-1 \times \text{threshold}$, is turned on simultaneously. The WL drives the sub-WL immediately, and the V_{BL} increases by accumulating the weights. Neuro-CIM achieves the difference (V_{DIFF}) between membrane potential and threshold by weighted-sum between all BLs, and the output spike occurs when the difference is larger than zero. To implement this, we compute positive part and negative part of V_{DIFF} separately and compare them to check whether positive part is larger than negative part. The two capacitor-DAC adder in Neuro-CIM generates positive partial sum (V_{POS}) and negative partial sum (V_{NEG}). V_{BLs} related to the positive part ($W[6:0]$) are transmitted to the positive adder, and the V_{BLs} related to the negative part (“−1” flags, $W[7] = \text{sign bit}$) are transmitted to the negative adder. The adder has capacitors to perform weighted sum with V_{BLs} for multibit operation. The input voltage of the capacitor is fixed to GND, and it is switched to the V_{BL} when performing an addition. At this time, the output voltage of the adder is changed by a value, which obtained by weighting the V_{BL} with the capacitor size. Also, we design 8-b capacitor-DAC adder by using two 4-b adder, which have $8C$, $4C$, $2C$, and $1C$ for area reduction. In 8-b mode, two adders are connected by the

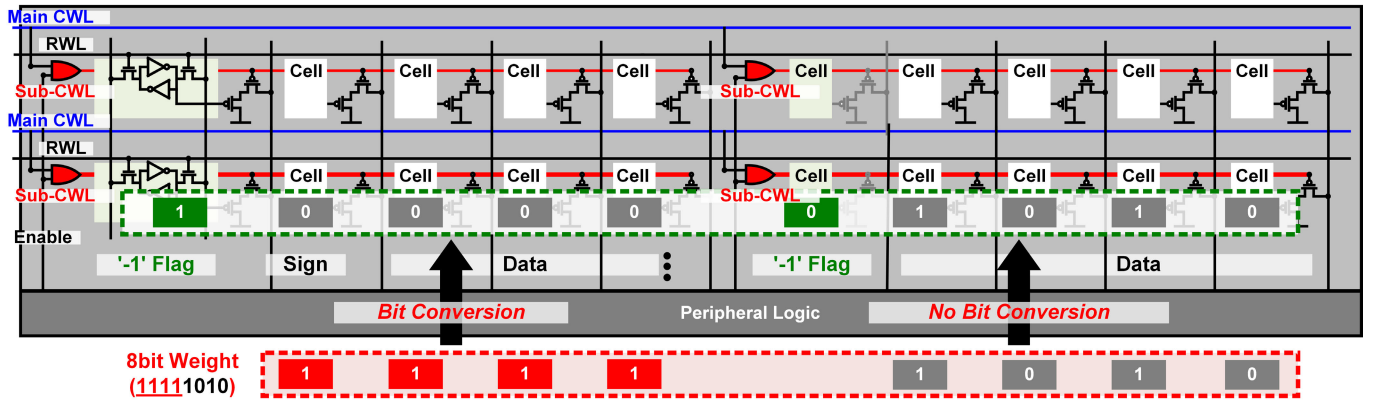


Fig. 9. Sign extended bits gating for 8-b weight mode.

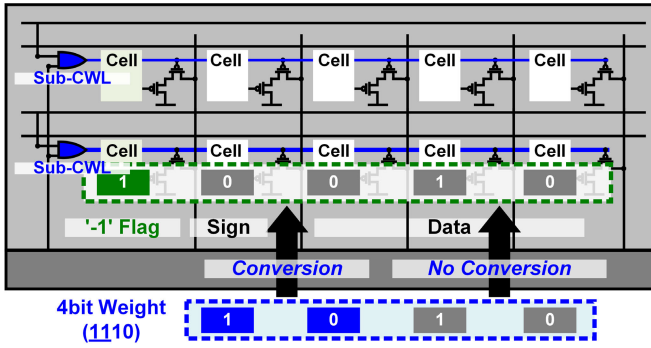


Fig. 10. Sign extended bits gating for 4-b weight mode.

bridge capacitor ($16/15C$), so that each VBL can be weighted by 2 to the n th power ($n = \text{bit position}$). The unit capacitor size (C) is 1 fF, and we design bridge capacitor by adding the additional metal plate. The unit capacitor is composed of three metal plates with an area of A corresponding to distance d , and through this, it has a capacitance of $2 \times A/d$. The bridge capacitor has a capacitance of $2 \times A/d + A/7.5d$ by placing another plate with an area of A at a distance of $7.5d$ from the unit capacitor. This is a value that is 16/15 times larger than the unit capacitor, and through this, we are able to implement the desired capacitor-DAC adder circuit in a small area. In 4-b mode, each 4-b adder independently performs 4-b operation. V_{BLs} associated with a positive value ($W[2:0]$) are transferred to the positive adder, and V_{BLs} associated with a negative value (“-1” flag, $W[3] = \text{sign bit}$) are transferred to the negative adder.

Then, the two voltages (V_{POS} and V_{NEG}) are transferred to the analog comparator, as shown in Fig. 12. When V_{POS} is larger than V_{NEG} , a comparator generates output spike, because the membrane potential exceeds the threshold. The output spike resets the V_{BLs} and increases the spike counter. If the V_{POS} is smaller than V_{NEG} , input spikes of next time step turn on the WL, and weight accumulation and firing are performed again. This operation is repeated until the output spike occurs. Fig. 12 shows the measurement waveforms of analog partial sums (V_{POS} and V_{NEG}) in 8-b weight mode. The weight accumulation is performed in each BL, and V_{BL} increases. Then, partial sums are generated by the capacitor-DAC multibit

adder. The red line shows the waveform of the positive partial sum (V_{POS}), and the black line shows the measured waveform of the negative partial sum (V_{NEG}). Measurements are performed at a frequency of 200 MHz, and 1.1-V conditions. In the first time step, the rising slope is larger than other time steps, because the $-1 \times \text{threshold}$ is added in partial sum. In other time steps, two voltages increase by only a spike pattern. In every time step, two partial-sum voltages are transmitted to the analog comparator for output spike generation. As shown at the 35 ns, when V_{POS} is larger than V_{NEG} , the output voltage of the analog comparator changes to high, and the spike counter increases. Then, both partial-sum voltages are reset to GND. In this process, only a comparison with the threshold is required. Therefore, Neuro-CIM has a great advantage in that it can process neural network without high-precision ADC. In order to measure output waveforms of the positive adder and negative adder operating before voltage folding, we put the voltage of each adder’s output node as the input of the buffer. Then, the output node of the buffer is connected to the pad, and the voltage output from the pad is measured with an oscilloscope. Accordingly, we are able to obtain the measured waveform, as shown in Fig. 12. Also, Fig. 13 shows a 1-bit dynamic comparator. First, when the enable signal is low, only the uppermost switch is turned off, all other switches are turned on, and COMP and COMP_B are initialized to 0. At this time, the switch in the middle connects both nodes to perform a more accurate initialization. When enable changes to high, comparison proceeds. Fig. 13 shows the operation for the case of $V_{IN1} > V_{IN2}$. Except for the uppermost switch, all switches for initialization are turned off, and the PMOS connected to V_{IN1} and V_{IN2} operates. At this time, the driving ability of the PMOS connected to the higher voltage becomes larger, and it causes a difference in the voltage of the cross-couple node. Eventually, COMP and COMP_B are completely torn into 0 and 1 by the cross-couple circuit.

C. Early Stopping Scheme

Fig. 14 shows that some neurons do not generate any output spikes during the entire time window due to small membrane potential. For low-power consumption, Neuro-CIM uses front part of the time window to predict the neurons with zero

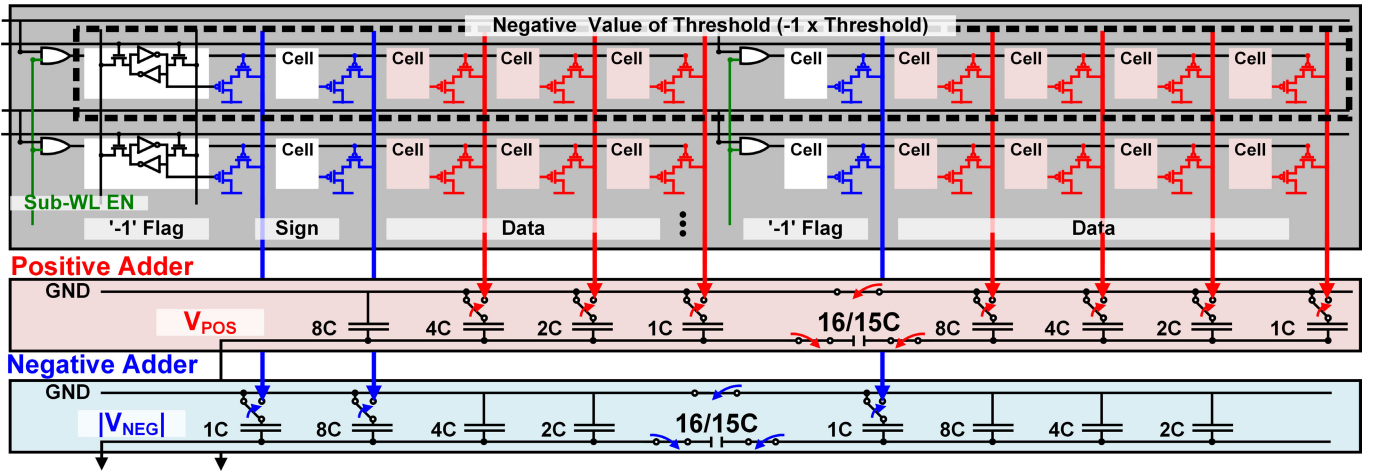


Fig. 11. Multibit operation in Neuro-CIM.

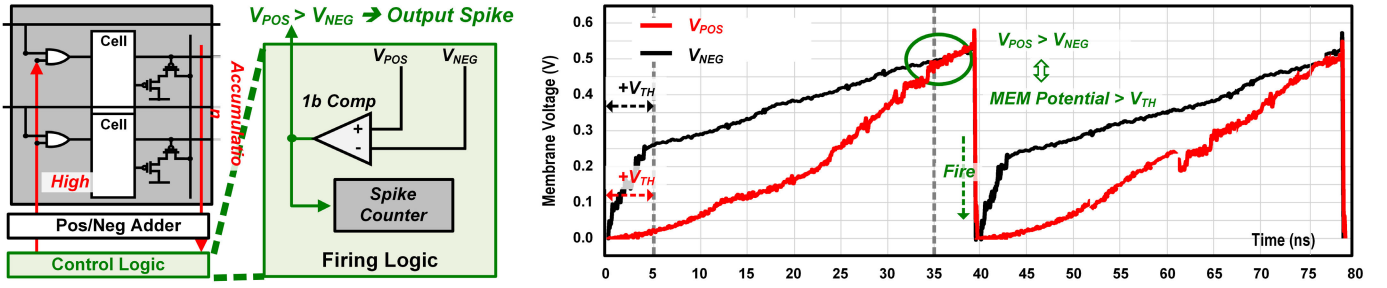


Fig. 12. ADC-less neuron operation with 1-b comparator and measured waveforms.

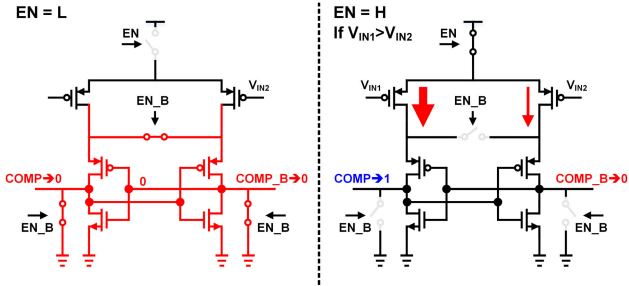


Fig. 13. Architecture of comparator in firing logic.

output spike, and it does not perform any further computations on the predicted neurons during the remaining time steps. The early stopping operation and measured waveform are described in Fig. 15. After an ADC-less neuron operation is performed during specific time steps, the ES_{EN} signal changes to high, so that the comparator in the early stopping logic can be operated for prediction. Neuro-CIM predicts the output spike to be zero when the accumulated membrane potential is less than a specific voltage (V_{ES}). First, the capacitor-based subtractor calculates the difference between V_{POS} and V_{NEG} from adder to generate the complete membrane potential. The generated voltage is transferred to the comparator and compared with an early stopping threshold (V_{ES}). If V_{ES} is larger than membrane potential, it is determined that the current neuron will not generate an output spike, and the output of the comparator is changed to low. The comparator

output is delivered to the sub-WL driver, and the output of sub-WL remains low for the remaining time steps. In a column in which sub-WL is disabled, the V_{BL} does not rise anymore even if a spike comes in, so the power consumption of the CIM is reduced. As a result, 50%–70% of neurons in each layer were stopped early within 1% of the accuracy loss, and power consumption decreases by $0.69\times$ (@CIFAR-10, ResNet-12). Moreover, 32% of neurons in whole network are terminated early with less than 1% accuracy loss for ImageNet inference (ResNet-50).

D. Analog Network With Voltage Folding-and-Count

Fig. 16 shows the concept of voltage folding-and-count in CIM architecture. During weight accumulation in BL, the V_{BL} is limited by $V_{BL,MAX}$, which cannot generate enough large V_{LSB} due to the range limitation. Therefore, the CIM performs inaccurate computing due to the limited range or small V_{LSB} . In this article, we fold the voltage of each macro and increase the folding count to represent the virtual range, as shown in the right of Fig. 15. By folding count, we can virtually increase the voltage range and guarantee enough large V_{LSB} for accurate comparison between potential and threshold.

Neuro-CIM integrates a voltage folding-and-count logic, as shown in Fig. 17, for low-power and high-accurate accumulation. Neuron generates two kinds of partial sums (V_{POS} , V_{NEG}), and both values continue to increase for several time steps. Each partial sum is used as an input voltage (V_{in}) for different folding logics. V_{in} is applied to N-MOSFET gates,

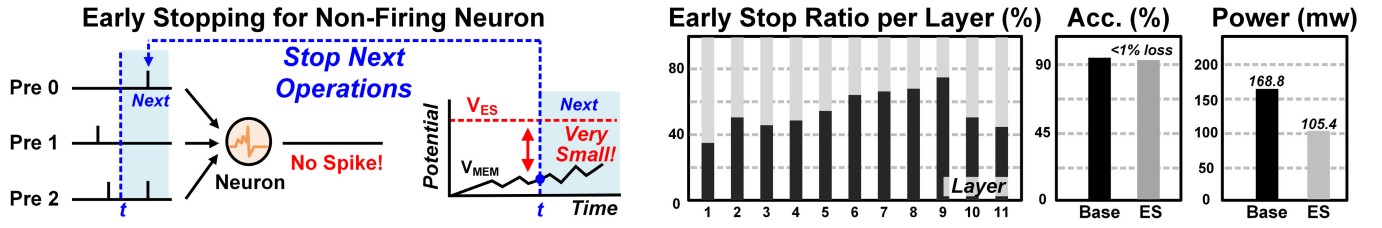


Fig. 14. Motivation of early stopping and performance.

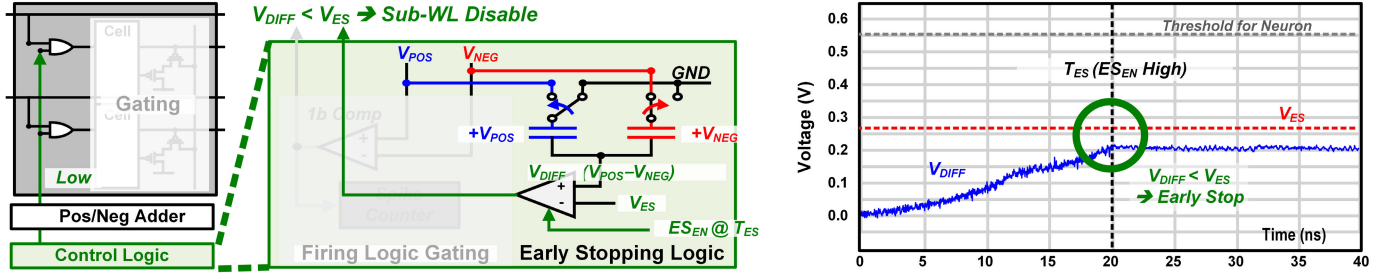


Fig. 15. Neuron operation with early stopping logic in Neuro-CIM.

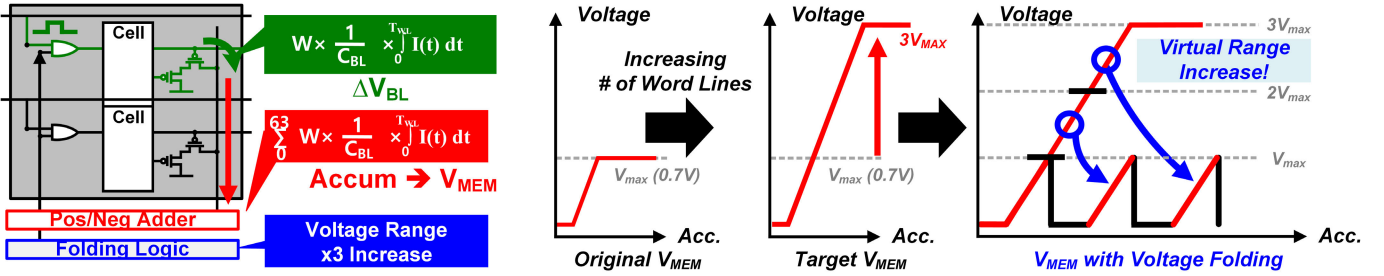


Fig. 16. Voltage range limitation of CIM and motivation of voltage folding-and-count.

and reference voltages (V_1 – V_4) are applied to another gates of the N-MOSFETs. Also, V_x and V_y are generated as outputs, as shown in Fig. 18. As V_{in} increases, V_x gradually decreases, and V_y increases gradually. When V_{in} reaches $(V_1 + V_2)/2$, V_x gradually increases, and V_y gradually decreases. Thereafter, when V_{in} reaches $(V_2 + V_3)/2$, the amounts of increase of V_x and V_y according to the increase of V_{in} have changed again. As a result, this circuit has three folding points, and it generates V_x and V_y with a phase difference of 180° .

Neuro-CIM can obtain an output waveform with a virtually continuously increasing range by selecting the ones with positive slopes among V_x and V_y . Only one of the two outputs is selected by the current folding counter value ($C[0]$) and transmitted to the analog comparator. The comparator changes the output voltage high when V_x or V_y reaches the maximum voltage V_{max} . When the output of the comparator is high, the value of the folding counter increases by 1, and the input voltage of the analog comparator to be compared with V_{max} is switched, because $C[0]$ is changed. Accordingly, a voltage having a positive rising slope is selected again among V_x and V_y , and it is compared with V_{max} . We utilize this voltage folding circuit for neuron operation, and the operation graph is shown at the bottom of Fig. 18. The V_{BL} increases every time step, and the partial sum (V_{NEG}) also increases continuously. When V_{NEG} reaches $V_{max} = (V_1 + V_2)/2$, the folding counter

increases by 1, and $V_{NEG,FOLD}$ is folded by initializing to GND. Afterward, when V_{NEG} increases and reaches $2 \times V_{max}$, $V_{NEG,FOLD}$ has reset to GND again, and the folding counter is increased by 1. As a result, V_{NEG} and V_{POS} are folded based on a specified maximum voltage by voltage folding circuit, and folding counts have been generated that record the virtual range of partial sum.

In addition, aggregation between macros is required for high network configurability. The V_{BL} of each macro corresponds to the accumulated weight value, and two methods can be largely utilized to perform input extension by adding the output values generated from different macros, as shown in Fig. 19. First, there is a method in which the V_{BL} generated in the cell array of each macro is transferred to the input of the ADC, and digital values are transferred to peripheral logic in specific macro to process the accumulation in the digital domain. However, this method has a very large power and area overhead, because ADC must be operated in every macro every time. Second, there is a method of charge sharing by connecting BLs in different macros to remove the ADC overhead. In this case, more cells are connected to one BL that means more charge is accumulated in the BL capacitor, and this means higher voltage range is required to maintain V_{LSB} . Therefore, we implement mixed-mode firing with voltage folding-and-count logic for high accuracy and low power.

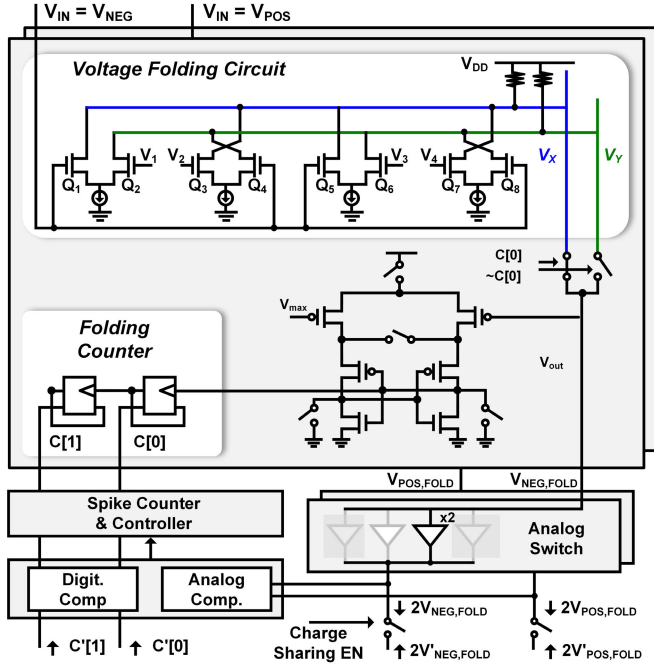


Fig. 17. Voltage folding-and-count logic.

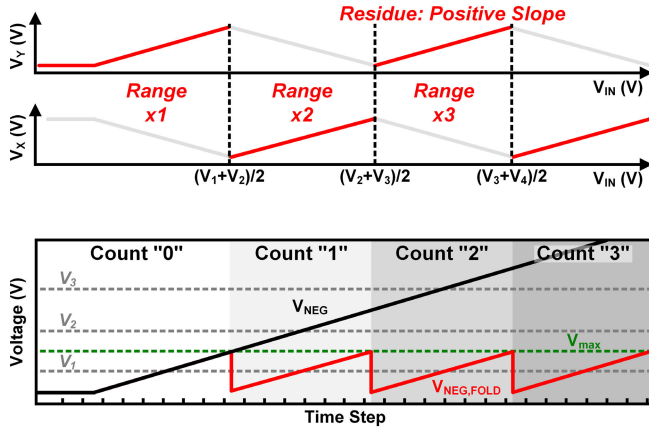


Fig. 18. Operation of voltage folding-and-count logic.

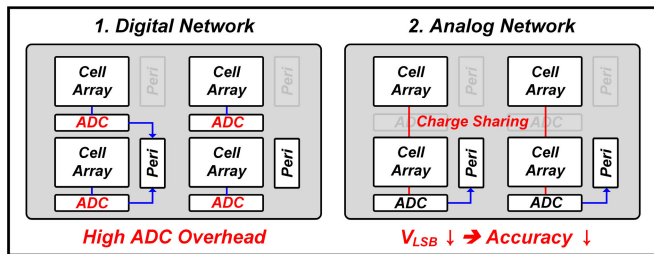


Fig. 19. Motivation of analog network with voltage folding-and-count.

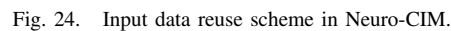
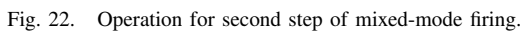
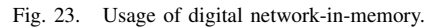
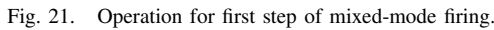
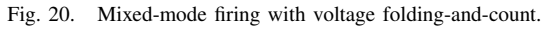
Fig. 20 shows the mixed-mode firing with two-step comparisons. This operation is performed by using both firing logic and the PGA. The first step is a coarse digital comparison, which uses the digital folding count to compare the virtual ranges. If the virtual range of the positive part is larger than the negative, the output spike fires, and the second step will be skipped. In another case, the second step (analog comparison)

is performed to compare values in the same virtual range, and the output spike fires when the positive voltage is larger than the negative. Fig. 21 shows the detail of mixed mode firing logic in Neuro-CIM for step 1. The Neuro-CIM can support both single-macro mode and multimacro mode firing for high network scalability. In single-macro firing mode, the positive and negative folding counts in single macro are transferred to the mixed-mode firing unit, and the coarse digital comparison is performed to generate an output spike. In multi-macro mode, different folding counts from different macros are transferred to a digital adder, and aggregation of each part is performed. Then, the digital comparison is performed. If the digital folding counts are the same, step 2 is performed to complete output spike firing. In step 2, also, firing logic (Fig. 22) can support two modes. In single mode, the positive and negative residue voltages in a single macro are transferred to 1-b analog comparator to generate an output spike. In multimacro mode, charge sharing between different voltages from different macros is performed for aggregation. In this case, we can maintain the V_{LSB} after charge sharing by controlling the gain of the residue amplifier for residue voltage. In the case of two macros are merged, the amplified voltages from two macros ($2 \times V_{NEG,FOLD}$ and $2 \times V'_{NEG,FOLD}$) are charge shared for aggregation. After charge sharing, the aggregation results ($V_{NEG,FOLD} + V'_{NEG,FOLD}$) are transferred to the analog comparator and complete the firing operation.

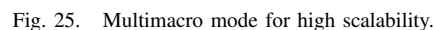
E. Digital Network-in-Memory and Data Reuse Scheme

Neuro-CIM has a reconfigurable structure to operate various neural networks. For this purpose, the digital network-in-memory architecture is implemented, as shown in Fig. 23, and it has three main functions. First, it enables output extension by transmitting the same data to multiple macros at the same time. In the case of the neural network, the input can be reused for multiple output channels. Each macro of Neuro-CIM can be assigned to different output channel operations, and an input spike generated from the one macro can be shared with other macros, which computes different output channels by digital network. Therefore, the network eliminates redundant operations for spike generation that occurs simultaneously in multiple macros. Second, it enables input extension by accumulating the outputs generated from different neurons. Neural network has very many input channels, and different input channels can be assigned to macros of Neuro-CIM. Third, it transfers the folding count of each macro to another macro for mixed-mode firing that enables multimacro aggregation. Therefore, Neuro-CIM can process various networks having different input channels or output channels by using the digital network-in-memory.

In addition, we implement input data reuse logic, as shown in Fig. 24. The spike encoder of each neuron has a dedicated input buffer, which is integrated to increase the energy efficiency by reusing the input data that can be used for multiple output channel operations. First, all data are stored in the input buffer, and the neuron operation is started for $layer_{l-1}$. When the operation of one neuron in a layer is finished, the generated result can be transmitted to another neuron in $layer_l$,



In summary, Neuro-CIM achieves high reconfigurability without high-cost ADC and accuracy degradation. An example of processing various configurations using the digital-analog network is described in Fig. 25. Since there are 64 WLs in each neuron, operations for 64 input channels can be assigned to the neuron. When the input channel of the layer is 128,



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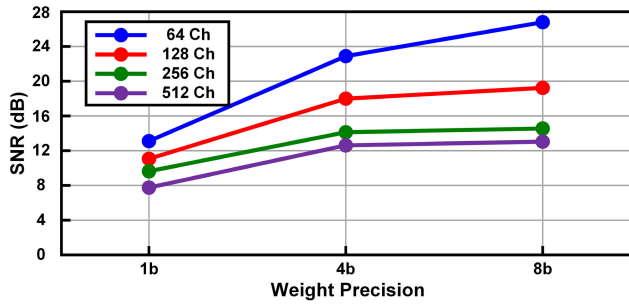
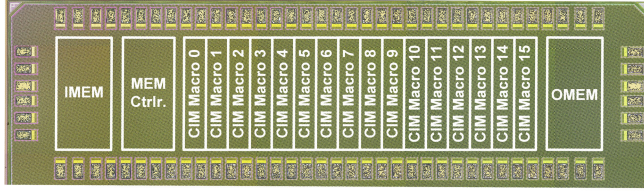


Fig. 26. SNR measurement results of each multimacro mode.



Specifications	
Technology	28nm
Die Area	3228μm × 900μm
CIM Macro Area	0.048 mm ²
CIM Macro Array Size	64×160
Total CIM Storage	20 KB
Digital SRAM	32 KB
Supply @ (Nominal)	1.1V
Frequency (Nominal)	200MHz
Input Precision	1b – 8b
Weight Precision	1b or 4b or 8b
Total Macro Power (mW)	15.84 (16 × 0.99) ³⁾ – 36.23 (16 × 2.26) ¹⁾
Total System Power (mW)	105.44 ³⁾ – 241.44 ¹⁾
Macro Energy Efficiency (TOPS/W)	723.47 ¹⁾ – 1034.57 ²⁾ (IN= 4b, W= 1b) 180.86 ¹⁾ – 413.83 ³⁾ (IN= 4b, W= 4b) 90.43 ¹⁾ – 206.91 ³⁾ (IN= 4b, W= 8b)
System Energy Efficiency (TOPS/W)	217.04 ¹⁾ – 310.37 ²⁾ (IN= 4b, W= 1b) 54.26 ¹⁾ – 124.15 ³⁾ (IN= 4b, W= 4b) 27.13 ¹⁾ – 62.07 ³⁾ (IN= 4b, W= 8b)
Network Architecture (ResNet-12)	L1: 64 CONV L5: 64 CONV L9: 256 CONV L2: 64 CONV L6: 128 CONV L10: 512 CONV L3: 64 CONV L7: 128 CONV L11: 512 CONV L4: 64 CONV L8: 256 CONV L12: 10 FC

1) Nominal Operation, 2) Operation w/ Early Stopping 3) Operation w/ Early Stopping & MSB Word Skipping

Fig. 27. Chip photograph and performance.

when the input channel of the layer is 256, four banks are connected through a hybrid network, and the gain of the analog switch increases, as the number of neurons increases. Applying the same method, Neuro-CIM can support up to 512 input channels. Fig. 26 shows the SNR measurement results for each mode. We use different multimacro modes for each layer and generate intermediate data by performing the operation of each layer on the actual chip. Based on this, the SNR is calculated under various weight precision conditions for each input channel. As a result of the measurement, the SNR tended to decrease as the number of channels increased. In other words, if more macros are grouped into one in an actual chip, the result of operation will be more affected by noise/variation.

IV. IMPLEMENTATION RESULTS

Fig. 27 shows the chip photograph and performance. Neuro-CIM is fabricated in 28-nm CMOS technology and occupies a 2.9-mm² area. The macro has 64 × 160 cells, and the area is

0.048 mm². It has digital SRAMs to store input and output, and a memory controller. Also, it has 16 CIM macros to process SNN. The total CIM macro size is 20 kB, and the 32-kB SRAM is integrated for input and output storing. The supply voltage is 1.1 V, and the operating frequency is 200 MHz. The supported input precision is 1–8 b, and we use the spike encoding, which was presented in [17]. The spike encoding method converts n -bit data into a spike train with a time-window length of 2^n . Neuro-CIM is designed to process 1–8-bit input, so it can process 2–256 time windows. Also, the supported weight precision is 1/4/8 b. In order to measure the performance of Neuro-CIM for various benchmark tests, we split the operation of the network and utilize external memory access to handle the operation of the entire network. First, some weights of layer 1 are loaded from external DRAM and loaded into the on-chip integrated CIM macros. Similarly, part of the input feature map of layer 1 is loaded from external memory and written to IMEM. Afterward, the result generated through CIM operation is stored in OMEM, and the output data are transferred to external DRAM. Since the weights stored in CIM can be reused after one convolution operation, Neuro-CIM receives other input feature map data not yet used for operation from external DRAM and performs CIM operation again. By repeating this, all operations for one layer are performed, and then, Neuro-CIM processes the next layer in the same way. This allows Neuro-CIM to perform operations on the entire network. The macro consumes 36.23 mW for CIFAR-10 classification with ResNet-12, and the power consumption can be reduced by 15.84 mW by using early stopping and the sign extended bits gating. In addition, the total system consumes 241.44 mW for CIFAR-10 classification with ResNet-12, and it can be reduced by 105.44 mW. Neuro-CIM achieves an energy efficiency of 124.15 tera operations per second/watt (TOPS/W) (input = 4 b and weight = 4 b) with 92.1% accuracy and 310.37 TOPS/W (input = 4 b and weight = 1 b) with 90.7% accuracy at 200 MHz, and 1.1-V conditions (@ResNet-12, CIFAR-10). In the case of CNN, one MAC is counted by two operations. In this work, integrate-and-fire (IF) operations (including zero/gated accumulations) during time window are equivalent with one MAC operation because of the CNN-to-SNN conversion. Therefore, we counted two operations when each neuron completes IF operations for all time window (# of operations = 2 × # of accumulations during time window/time window). In summary, the operations of the converted SNN are the same as that of CNNs with the same configuration. The system power defined in Fig. 27 is the total power consumed by the on chip, including CIM macro power, IMEM–OMEM power, memory controller, and digital/analog network power. Since the previously proposed processors [2], [3], [4] did not report off-chip power, only system power including on-chip power was reported in this article for a fair comparison. The macro shows 1.03-POPS/W (peta operations per second/watt) energy efficiency at 4-b input and 1-b weight and 413.83-TOPS/W energy efficiency at 4-b input and 4-b weight. ResNet-12 has three convolutional layers, four ResNet blocks, and one fully connected layer. Each convolutional layer has 64 filters of size 3 × 3 and stride 1, followed by batch normalization and ReLU activation. Each ResNet block

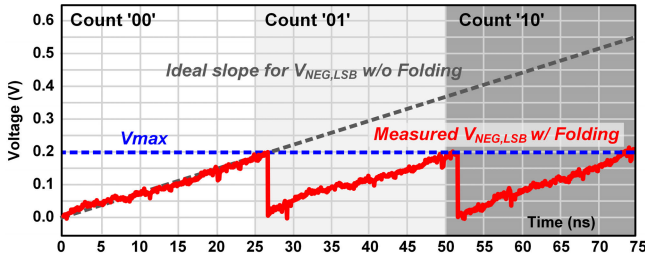


Fig. 28. Measurement waveforms of voltage folding circuit.

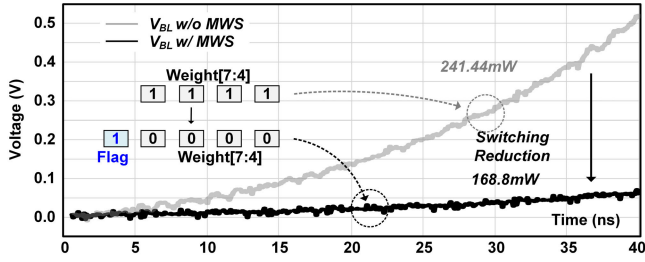


Fig. 29. Measured waveforms for sign extended bits gating.

consists of two convolutional layers and shortcut connection, and each convolutional layer in ResNet block has 64 filters of size 3×3 and stride 1 with batch normalization and ReLU activation. One fully connected layer has ten output units corresponding to the ten classes in the CIFAR-10 dataset.

Fig. 28 shows the measurement waveform of voltage folding under 1.1 V and 200 MHz. The red line is $V_{NEG,FOLD}$, which is the folded negative partial sum. The gray line is the ideal waveform of $V_{NEG,FOLD}$ when there is no folding, and the blue line is the maximum voltage used for voltage folding. WLs are selectively enabled by the spike train with a constant spike ratio during 75 ns. As the time-step progresses, the V_{BL} s and the partial-sum voltages increase continuously, and partial-sum voltages are transferred to the voltage folding circuit. When the partial sum reaches the maximum voltage, voltage folding is performed, as shown in the graph of Fig. 28. Accordingly, $V_{NEG,FOLD}$ is initialized to GND, and the folding count is increased. After that, when the time step continues and the $V_{NEG,FOLD}$ reaches the maximum voltage again, voltage folding is performed again, and the count value increases. After that, the corresponding operation is repeated until the time window ends.

Fig. 29 shows the measurement waveform of sign extended bits gating with 8-b weight. The gray line is the V_{BL} waveform of the column where the highest bit of the weight ($W[7]$) is stored, and the waveform is measured when the proposed feature is not utilized. The black line is the V_{BL} waveform for the same column when the proposed feature is applied. Without sign extended bits gating, a lot of “1”s are stored in the cells, because there are a lot of consecutive “1”s in the upper 4 bit. Therefore, a very large amount of charge is accumulated in the BL capacitor, and the V_{BL} rises very much. When sign extended bits gating is applied, consecutive “1”s in the upper 4 bit are removed, and “1” is written in flag cell. Therefore, many cells in the columns corresponding to the upper bit store “0.” When the neuron operation starts, the rise

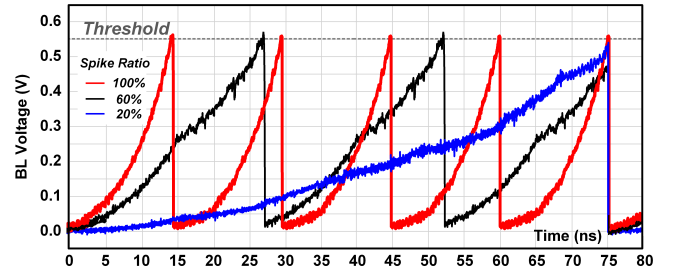


Fig. 30. Measured waveforms of BL operation at 1-b weight mode.

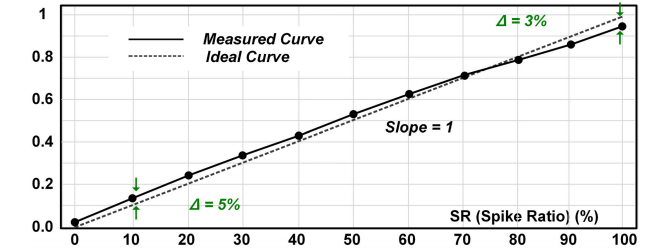


Fig. 31. Output and input spike ratio linearity of Neuro-CIM.

of the V_{BL} becomes very small, as shown in Fig. 29. As a result, the power consumption of Neuro-CIM for processing ResNet is reduced by 30.1%.

Fig. 30 shows the V_{BL} measurement waveform at 1-b weight mode. In this case, the V_{BL} is directly transferred to the comparator without multibit adder operation. Neuron operation is performed at 200-MHz frequency and 1.1-V conditions, and the measurement results are achieved at different spike ratio conditions. The time step of SNN is 5 ns, which is one clock cycle, and the WL of the row where the spike occurs is turned on for 1 ns at the rising edge of the clock. Thereafter, the V_{BL} is determined based on the ratio of “1” stored among the WL-enabled cells in each column. The blue line shows the V_{BL} when the spike ratio is 20%. At 75 ns, the V_{BL} reaches the neuron’s threshold voltage, and the comparator can detect it. So, the output spike is generated, and the V_{BL} is reset to GND. The black line shows the V_{BL} when the spike ratio is 60%, and three times as many output spikes are generated during the same time compared with the 20% spike ratio. The red line is a V_{BL} measurement waveform when the spike ratio is 100%, and the proposed processor generates five times more output spikes than when the spike ratio is 20%.

Fig. 31 shows the normalized output spike ratio according to the spike ratio. The dotted line is an ideal curve in which the output spike generation period decreases linearly, as the input spike ratio increases. Also, we measure the time it takes for the spike generation while increasing the spike ratio by 10%, and the black line is the measurement waveform. The measurement result shows that the output spike firing ratio is linearly proportional to the input spike ratio with less than 5% linearity error.

Table I shows the comparison table for Neuro-CIM. The previous works [1], [2], [3], [4], [5], [6] use an ADC with 4–5-bit precision, and [2] uses an ADC with adjustable precision, whereas the proposed processor can operate without high-cost ADC. Therefore, the area efficiency of the macro

TABLE I
COMPARISON TABLE

	VLSI21 [1]	ISSCC21 [2]	ISSCC21 [4]	ISSCC21 [5]	ISSCC20 [6]	ISSCC21 [21]	This Work
Tech. (nm)	28	65	28	65	7	16	28
System Area (mm ²)	-	12	-	3.3	0.0032	25	2.9
Size of Macro (bit)	64×64	128×128	384K	64×32	4K	1152×256	64×160
Area of Macro (mm ²)	0.51	0.43	-	0.05	-	-	0.048
Input Precision	5	2/4/6/8	4/8	4/8	4	1-8	1-8
Weight Precision	1	1-8	4/8	4/8	4	1-8	1/4/8
Output Precision	8	0/2/4	5	5	4	8	8
Area Efficiency of Macro (TOPS/mm ²)	2.4 (l: 5b, W: 1b)	2.2 (l: 2b, W: 1b)	-	8.8 (l: 4b, W: 4b)	116.4 (l: 4b, W: 4b)	-	546.1 (l: 4b, W: 1b) 136.5 (l: 4b, W: 4b)
Peak Energy Efficiency of Macro (TOPS/W)	1159.2 (l: 5b, W: 1b)	370.4 (l: 2b, W: 1b)	60.3 – 94.3 (l: 4b, W: 4b)	102.2 – 216.8 (l: 4b, W: 4b)	351 (l: 4b, W: 4b)	-	1034.6 (l: 4b, W: 1b) 413.8 (l: 4b, W: 4b)
Peak Energy Efficiency of System (TOPS/W)	-	75.9 (l: 2b, W: 1b)	-	32.5 – 44.7 (l: 4b, W: 4b)	-	121 (1b normalize)	310.4 (l: 4b, W: 1b) 124.2 (l: 4b, W: 4b)
Energy / Classification (CIFAR-10)	1.73 μ J (91.1 %)	- (91.71 %)	-	- (91.2 %)	-	19.4 μ J (91.51%)	0.72 μ J (92.8 %)
Energy / Classification (ImageNet)	-	- (66.1 %)	-	-	-	334 μ J (73.08%) @ 4b input, weight	372.2 μ J (72.5 %) @ 8b input, weight

and system is much higher than the previous processors. Yue et al. [2] integrate $1.6\times$ more cells in the macro than the proposed processor. However, the macro in [2] occupies an area $8.96\times$ larger than Neuro-CIM. Chen et al. [4] occupy an area similar to the proposed macro, but integrates $5\times$ fewer cells. Moreover, the area efficiency increases by $4.69\times$ compared with the previous state of the art [6]. In addition, the proposed processor can perform early stopping by predicting the neurons that do not generate any output spike, but previous works [1], [2], [3], [4], [5], [6] cannot support the prediction for power reduction. Moreover, Neuro-CIM is the only CIM processor that supports the sign extended bits gating, which can significantly reduce power consumption during multibit operation. Because of power-reducing features, Neuro-CIM achieves state-of-the-art energy per classification. The proposed processor decreases the energy consumption for CIFAR-10 classification by 58.4% compared with the previous state of the art [1], and Neuro-CIM also consumes only 372.2 μ J for ImageNet classification (one frame). Neuro-CIM shows a latency of 3.28 ms when performing CIFAR-10 classification under 4-bit input/weight conditions using ResNet-12. When using VGG-16 under the same conditions, the latency is 9.9 ms. Also, when Neuro-CIM performs ImageNet classification with ResNet-50, it shows a latency of 49.8 ms. For ImageNet classification, we use ResNet-50, and the size of the input image is 220×220 .

Fig. 32 shows the measurement system of Neuro-CIM. Neuro-CIM board is connected to the FPGA system board, dc power analyzer, and computer. Neuro-CIM receives the input image from the host computer, and the computation result of Neuro-CIM is transferred to the host by the FPGA system. As a result, the top-5 result is displayed on the monitor of the computer, and the results are used to measure computing accuracy. Neuro-CIM achieves 92.1% accuracy for the CIFAR-10 (4-b input and 4-b weight) and 72.5% accuracy for the ImageNet (6-b input and 8-b weight). In addition, Neuro-CIM achieves an energy efficiency of 124.15 TOPS/W for the CIFAR-10 with ResNet-12, which requires 0.06 GOP/frame (giga operations per frame). Also, Neuro-CIM can achieve 117.98 and 103.06 TOPS/W for VGG-16 and ResNet-18, respectively (CIFAR-10 classification with 4-b input and 4-b

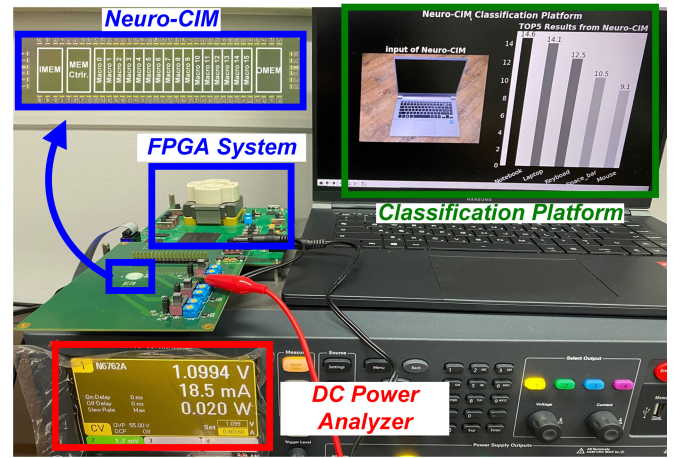


Fig. 32. Measurement system of Neuro-CIM.

weight). VGG-16 requires 0.63 GOP/frame and ResNet-18 requires 0.21 GOP/frame for CIFAR-10 task. Moreover, it achieves an energy efficiency of 37.24, 32.61, and 15.1 TOPS/W for VGG-16, ResNet-18, and MobileNet, respectively (ImageNet classification with 6-b input and 8-b weight). VGG-16 requires 29.9 GOP/frame, ResNet-18 requires 6.9 GOP/frame, and MobileNet requires 0.56 GOP/frame. The macro shows 1.03-POPS/W peak energy efficiency at 4-b input and 1-b weight and 413.83-TOPS/W peak energy efficiency at 4-b input and 4-b weight.

Neuro-CIM shows a difference in utilization when processing a convolutional layer and a fully connected layer. In the case of a convolutional layer, the weights stored in CIM can be reused after performing one convolution. Therefore, the time required to write weights to CIM is short, and the impact of utilization drop due to data movement is very small. However, in the case of a fully connected layer, the weights cannot be reused again after being used for calculation once. Therefore, since weights must be newly written to CIM each time, computation utilization decreases accordingly. As a result, when performing CIFAR-10 classification in Neuro-CIM, ResNet-12 with a small number of fully connected layers has a latency of 3.28 ms, but VGG-16 with a high ratio of fully connected layers shows a latency of 9.9 ms.

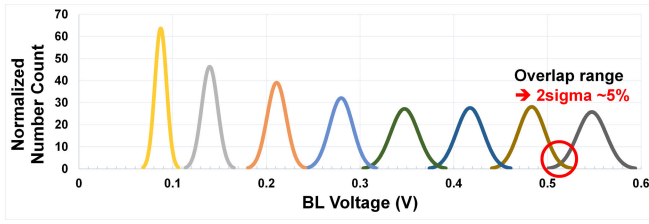


Fig. 33. Monte Carlo results of Neuro-CIM.

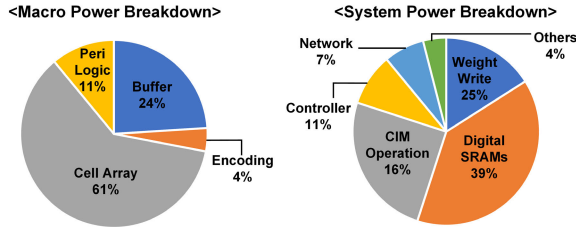


Fig. 34. System power breakdown and macro power breakdown.

Our measurement is performed with CNN-to-SNN conversion with three steps. First, to find the weight of each synapse of the SNN, the CNN is first trained through backpropagation. After that, the trained weight is shared to each synapse of the SNN without additional training. Second, spike encoding is used to convert input data into spike domain. In this work, rate encoding is used, where information is encoded as the firing rate of neurons over time. Each input data are compared with a random value for a specified time-window size. When the data value is greater than the random value, a spike is generated, and through this, an input spike train with a length equal to the time-window size is created. Third, the threshold of each neuron of the SNN is found using the generated input spike. In this process, there is a power/accuracy overhead. As the threshold increases, the frequency at which each neuron fires decreases, resulting in a decrease in the number of operations, resulting in a decrease in power. However, loss of information results in a decrease in accuracy. Therefore, the threshold must be set to achieve the minimum power according to the target accuracy.

Moreover, noise, nonlinearity, and process, voltage, temperature (PVT) variation can reduce accuracy during measurement. In particular, in processing-in-memory (PIM), the PVT variation between many SRAM cells is the biggest factor that reduces calculation accuracy. Neuro-CIM performs accumulation on BL, and the voltage of BL corresponds to the operation result. In this case, if we want to accumulate many values at once, we need to integrate a large BL capacitor to have a small V_{LSB} , and the influence of noise increases due to small V_{LSB} . However, in the case of SNN, the number of spikes firing in one time step is very small due to the event-driven characteristics. In addition, since it is only necessary to determine whether the neuron's membrane potential has exceeded the threshold for each time step, the range of maximum accumulation result can be reduced. Therefore, we can reduce the size of the BL capacitor for the accumulation and use a large V_{LSB} , so that it was not vulnerable to noise.

Fig. 33 shows the Monte Carlo simulation result. When the hardware has sufficient V_{LSB} , an error of up to two

sigma occurs. However, when software simulation is actually performed based on this information, only 0.2% of software-accuracy loss occurs in CIFAR-10, which is negligible. In addition to the BL accumulation process, accuracy may decrease due to nonlinearity and variation in other analog circuit parts. This problem can be solved by using the training method proposed in [6] and [19]. The relationship between input and output (transfer function) can be obtained from hardware, and based on this, an SNN model with nonlinear modeling applied can be created. Afterward, the modeled network can be retrained to improve accuracy in actual hardware. Due to the noise and variation of the voltage folding circuit and other analog circuits, the hardware accuracy (86.4%) for CIFAR-10 classification is reduced by 7.1% compared with the software accuracy (93.5%) (@4-bit weight, ResNet-12). However, with the SNN model fine-tuned through the mentioned technique, Neuro-CIM can achieve 92.8% hardware accuracy. In this fine-tuning method, each chip has a different transfer function, because each chip has a different variation. If the SNN retrained for one chip is loaded into another chip and the calculation is performed, there is a slight decrease in accuracy. Therefore, in order to obtain optimal accuracy for each chip, retraining should be performed separately. Finally, Fig. 34 shows the system and macro power breakdown of Neuro-CIM.

V. DISCUSSION

In this section, we discuss about how can Neuro-CIM can achieve higher energy efficiency compared with previous CIM processors [1], [2], [3], [4], [5], [6]. Previous CIM processors consume lots of power due to redundant V_{BL} switching by many consecutive “1’s” in upper bits. The proposed processor uses flag BL and sign extended bits gating to reduce the BL voltage of columns corresponding to upper bits. Neuro-CIM converts consecutive “1’s” to zero and records this information to flag BL, and the V_{BL} for the upper bit is greatly reduced. As a result, the proposed processor reduces the power consumption for CIFAR-10 classification by 30.1% by sign extended bits gating. The proposed method can be compared with the sign magnitude operation. In the case of sign magnitude [20], as in the scheme proposed in this work, the number of 0’s in the weight data can be increased. However, when the sign magnitude method is used, a problem arises in the process of accumulation within the BL, unlike the 2’s complement method. Assuming that A is positive and B is negative, the 2’s complement method flips each bit of B and adds 1 to the converted B . When adding A and B in PIM, A and B expressed in 2’s complement are first stored in the cell. Then, accumulation between each bit of A and B occurs in each BL, and the final addition result can be created through multibit aggregation. However, in the case of sign magnitude, in the process of adding two different numbers, subtraction or addition must be performed according to the sign, rather than simply adding the magnitudes of the two numbers to each other. In the process of adding A and B , since the signs of the two data are different, subtraction must be performed for each bit. However, in PIM, only addition between multiple data in one BL can be performed. Therefore, it is difficult to

directly implement the sign magnitude calculation method in PIM. In addition, the previous processors cannot support early termination by predicting the zero output value. However, the proposed processor supports early stopping that the neuron operation for zero output spike is early terminated. As a result, Neuro-CIM reduces the power consumption for ResNet-12 computation by 37.5%.

Moreover, previous processors have a large overhead to achieve reconfigurability. Digital network has high ADC overhead, because every macro should operate ADC every time to aggregate the partial sums from different macros. Analog network has a large accuracy loss, because the voltage scale decreases after charge sharing. However, we propose an aggregation method with digital and analog networks. Voltage folding-and-count is used to increase the virtual range, and mixed-mode firing is performed to reduce power consumption and maintain the voltage scale after charge sharing.

VI. CONCLUSION

Neuro-CIM is proposed for energy-efficient neural network processing by using neuromorphic computing. Conventional CIM processors [1], [2], [3], [4], [5], [6] use high-cost ADC to generate output value. However, the proposed processor adopts IF neuron to achieve high energy efficiency. The reason of adopting IF neuron instead of leaky IF neuron is that most CNN-to-SNN conversion algorithms use IF neuron, because CNN does not have factors that decay over time, such as leaky IF neurons. So, Neuro-CIM can eliminate the high-cost ADC, because IF neuron of SNN requires only a comparison between the membrane potential and the threshold. As a result, Neuro-CIM with an energy efficiency of 310.37 TOPS/W and an area efficiency of 546.1 TOPS/mm² is proposed in this article for high energy-efficient neural network processing.

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